SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5113

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:100**

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Siva Kumar M (192521057) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Sivakumar m
Signature of the student:

Q. No	*Understanding of Concepts (25)*	*Experimental Design (30)*	*Use of Tools (20)*	*Analysis of Results (15)*	*Documentation (10)*	*Total Score (out of 100)*
1	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
2	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
3	*/ 5*	*/6*	*/ 4*	*/ 3*	*/ 2*	
4	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
5	*/ 5*	*/ 6*	*/ 4*	*/ 3*	*/ 2*	
OVERALL RUBRICS VALUE						
	/ 5	*/ 5*	*/ 4*	*/ 4*	*/ 2*	
	/ 25	*/ 30*	*/ 20*	*/ 15*	*/ 10*	*/ 100*

APPLICATION- BASED QUESTIONS

Q1. Scenario: Event Communication Security:

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

Given

- Plaintext: MEET AT NOON
- Shift value = 4
- Encryption Technique: Caesar Cipher

Step 1: Caesar Cipher Formula

Encryption:

$$C = (P + 4) mod 26$$

Where:

- P = Plaintext letter
- C = Ciphertext letter

Step 2: Encrypt Each Letter

Plaintext	Position	Shift +4	Ciphertext
M	12	16	Q
E	4	8	I
E	4	8	I
T	19	23	X
-	-	-	-
A	0	4	E
T	19	23	X
-	-	-	-
N	13	17	R
O	14	18	S
O	14	18	S
N	13	17	R

Ciphertext
QIIX EX RSSR

Step 3: Decryption

The receiver already knows the shift is 4.
Decryption formula:

$$P = (C - 4) mod 26$$

Example:
Q → M
I → E
I → E
X → T
Repeating for every letter gives
MEET AT NOON

Analysis

- Caesar cipher shifts every letter by a fixed amount.
- It is easy to implement.
- Only 26 possible keys exist.
- Therefore, it can be broken easily using brute-force attacks.

Result

Original Message
MEET AT NOON
↓
Encrypted
QIIX EX RSSR
↓
Decrypted
MEET AT NOON

Q2. Scenario: Intercepted Suspicious Message :

A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
Deduce the most likely original message. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

Given Ciphertext

GSRH RH Z HVXIVG

Observation

The ciphertext contains repeated words.

- RH appears twice.
- One-letter word = Z

This matches the characteristics of the **Atbash Cipher**.
In Atbash,

Plain	Cipher
A	Z
B	Y
C	X
D	W
E	V
F	U
G	T
H	S
I	R
J	Q
K	P
L	O
M	N
N	M
O	L
P	K
Q	J
R	I
S	H
T	G
U	F
V	E

Plain	Cipher
W	D
X	C
Y	B
Z	A

Decrypt Letter-by-Letter

Cipher	Plain
G	T
S	H
R	I
H	S

Original Message
THIS IS A SECRET

Justification

1. Repeated word RH becomes IS.
2. Single-letter word Z becomes A.
3. Letter frequency perfectly matches English.
4. Atbash is a substitution cipher with reversed alphabets.

Result

Ciphertext
GSRH RH Z HVXIVG
↓
Plaintext
THIS IS A SECRET

Q3. **Scenario: Secure Messaging System**:

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality. Demonstrate how the encryption process transforms the message step-by-step. Explain how the repeating keyword influences each character of the ciphertext.

Plaintext
HELLO

Keyword

KEY

Step 1: Repeat Keyword

Plain text : H E L L O

Keyword : K E Y K E

Step 2: Alphabet Positions

Letter	Number
H	7
E	4
L	11
L	11
O	14

Keyword

Letter	Number
K	10
E	4
Y	24
K	10
E	4

Step 3: Encryption
Formula

$$C = (P + K) mod 26$$

Plain	P	Key	K	Sum	Cipher
H	7	K	10	17	R
E	4	E	4	8	I
L	11	Y	24	35→9	J
L	11	K	10	21	V
O	14	E	4	18	S

Ciphertext
RIJVS

Effect of Repeating Keyword
The keyword repeats continuously.
KEYKEYKEY...
Each plaintext letter is shifted by a different amount.
Example
H shifted by K
E shifted by E
L shifted by Y
Therefore,
Same letters may encrypt differently depending on keyword position.
This makes Vigenère much stronger than Caesar Cipher.

Result
Plaintext
HELLO
↓
Keyword
KEY
↓
Ciphertext
RIJVS

Q4 Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection. Analyze how investigators could detect the presence of hidden data within the image. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

Problem

Investigators suspect that confidential information is hidden inside a company logo image.

Detecting Hidden Data

Method 1: LSB (Least Significant Bit) Analysis

Many steganography techniques modify only the least significant bits of pixels.
Investigators can compare pixel values.
If unusual LSB patterns exist, hidden data may be present.

Method 2: Statistical Analysis

Investigators examine

- Histogram
- Pixel distribution
- Entropy
- Noise patterns

Abnormal statistical properties indicate embedded data.

Method 3: File Size Comparison

If

- Original logo = 300 KB
- Suspicious logo = 450 KB

then additional hidden information may exist.

Method 4: Steganalysis Tools

Common tools include

- StegExpose
- Stegdetect
- OpenStego
- zsteg
- Binwalk

These tools inspect images for hidden payloads.

Two Methods to Improve Security

1. Encrypt Before Hiding
Plaintext
↓
AES Encryption
↓
Ciphertext
↓
Hide inside image
Even if extracted, attackers cannot understand the content without the encryption key.

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2. Random Pixel Embedding

Instead of storing data sequentially,
hide it in randomly selected pixels using a secret key.
Advantages

- Harder to detect.
- Resistant to statistical attacks.

Result Analysis

Combining encryption with steganography provides two levels of security:

1. Hidden communication.
2. Protected content even if detected.

Q5. Scenario: Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEEFEAOCAIVDEN”.

a. Determine the most probable original message by reversing the encryption process.
b. Critically evaluate why such a method is vulnerable in modern communication systems.
c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Given Ciphertext

WECRLTEERDSOEEFEAOCAIVDEN
This is encrypted using a **Rail Fence Cipher** with **3 rails**.

(a) Decryption

Step 1: Rail Pattern

Rail 1 : W . . . E . . . C . . . R . . . L . . . T . . . E
Rail 2 : . E . R . D . S . O . E . E . F . E . A . O . C .
Rail 3 : . . C . . . R . . . I . . . V . . . D . . . N

Reading in zigzag order reconstructs the message:
WE ARE DISCOVERED FLEE AT ONCE

Without spaces:
WEAREDISCOVEREDFLEEATONCE

Original Message
WE ARE DISCOVERED FLEE AT ONCE

(b) Why Rail Fence Cipher is Vulnerable
1. No Key Complexity
Only the number of rails acts as the key.
Usually only 2–10 possibilities exist.

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2. Easily Broken
Attackers try different rail values.
The correct English sentence quickly appears.

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3. Letter Frequency Preserved
The letters themselves are unchanged.
Frequency analysis is still possible.

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4. Weak Against Modern Computing
Modern computers test every possible rail arrangement within milliseconds.

(b) Practical Enhancement

Recommendation
Encrypt the message using **AES (Advanced Encryption Standard)** before transmission.
Advantages:

- Uses a strong secret key (128, 192, or 256 bits).
- Resistant to brute-force attacks.
- Widely used in banking, military, and secure communication.
- Provides strong confidentiality.

Alternatively, if a classical enhancement is required, combine **Rail Fence + Vigenère Cipher** to increase security.

Final Result

- **Original Ciphertext:** WECRLTEERDSOEEFEAOCAIVDEN
- **Recovered Plaintext: WE ARE DISCOVERED FLEE AT ONCE**
- **Weakness:** Easy to break because only letter positions change.
- **Improvement:** Use AES (or combine transposition with a substitution cipher such as Vigenère)